

Examen Práctico 1ª Evaluación

Caso de Exfiltración de Datos en Call4U S.L.

La empresa Call4U S.L., dedicada a la gestión comercial mediante tareas de televenta, detecta irregularidades en la información de clientes. Se sospecha que un trabajador del departamento de teleoperadores ha exfiltrado datos personales sensibles. El 29/11/2025, el departamento de IT de Call4U S.L. detectó actividad anómala en una de las estaciones de trabajo. Se sospecha que uno de los empleados que usa ese equipo habría extraído información confidencial del archivo clientes.xlsx, almacenado en el recurso compartido del servidor interno. Para ello, el departamento de IT llama al analista forense y le da acceso al equipo sospechoso con la sesión iniciada con el usuario Admin y le da tan solo unos minutos para que recopile datos, ya que las llamadas no pueden cesar. Con la información recopilada debe demostrar quien ha sido el responsable y el método empleado en la exfiltración.

1. Realizacion del triaje

```
C:\Users\Admin\Desktop\kape>kape.exe --tsource C: --tdest C:\Users\Admin\Desktop\traje --tflus -target KapeTriage --debu
```

Dentro de la máquina de Admin ejecutamos por línea de comandos Kape, el cual nos realizará un triaje del equipo entero para ser analizado posteriormente

```
Administrador: Símbolo del sistema
5604121 | Source modified: 2025-12-03 15:04:58.1075460 | Source last accessed: 2025-12-03 15:04:58.1075460
Copied deferred file C:\Users\Admin\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.jfm to C:\Users\Admin\Desktop\
traje\C\Users\Admin\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.jfm. Hashing source file...
Deferred file details | Source SHA-1: 7403B55FD8CE0E8B32B7B06B939936878B98CA50 | Source created: 2025-11-26 17:43:45.
5528386 | Source modified: 2025-11-26 17:56:20.8061762 | Source last accessed: 2025-11-26 17:56:20.8061762
Creating directory C:\Users\Admin\Desktop\traje\C\Users\Admin\AppData\Local\ConnectedDevicesPlatform\L.Admin
Copied deferred file C:\Users\Admin\AppData\Local\ConnectedDevicesPlatform\L.Admin\ActivitiesCache.db to C:\Users\Admin
\Desktop\traje\C\Users\Admin\AppData\Local\ConnectedDevicesPlatform\L.Admin\ActivitiesCache.db. Hashing source file...
Deferred file details | Source SHA-1: 1C5D9FB87C690EE1507EE8AEEA5C16FFD356B8F7 | Source created: 2025-11-26 17:43:54.
1123987 | Source modified: 2025-11-26 17:56:21.0747417 | Source last accessed: 2025-11-26 17:56:21.0747417
Copied deferred file C:\Users\Admin\AppData\Local\ConnectedDevicesPlatform\L.Admin\ActivitiesCache.db-shm to C:\Users\A
dmin\Desktop\traje\C\Users\Admin\AppData\Local\ConnectedDevicesPlatform\L.Admin\ActivitiesCache.db-shm. Hashing source f
ile...
Deferred file details | Source SHA-1: 389FBA53344102D1BAC46D66510E897CDDCECFCC | Source created: 2025-11-26 17:43:54.
1382718 | Source modified: 2025-12-03 15:04:58.3701959 | Source last accessed: 2025-12-03 15:04:58.3701959
Copied deferred file C:\Users\Admin\AppData\Local\ConnectedDevicesPlatform\L.Admin\ActivitiesCache.db-wal to C:\Users\A
dmin\Desktop\traje\C\Users\Admin\AppData\Local\ConnectedDevicesPlatform\L.Admin\ActivitiesCache.db-wal. Hashing source f
ile...
Deferred file details | Source SHA-1: 6241740471A069CEAD98CE28D2B975128BF358C1 | Source created: 2025-11-26 17:43:54.
1382718 | Source modified: 2025-12-03 15:11:51.5287448 | Source last accessed: 2025-12-03 15:11:51.5287448
Flushing writers and streams...

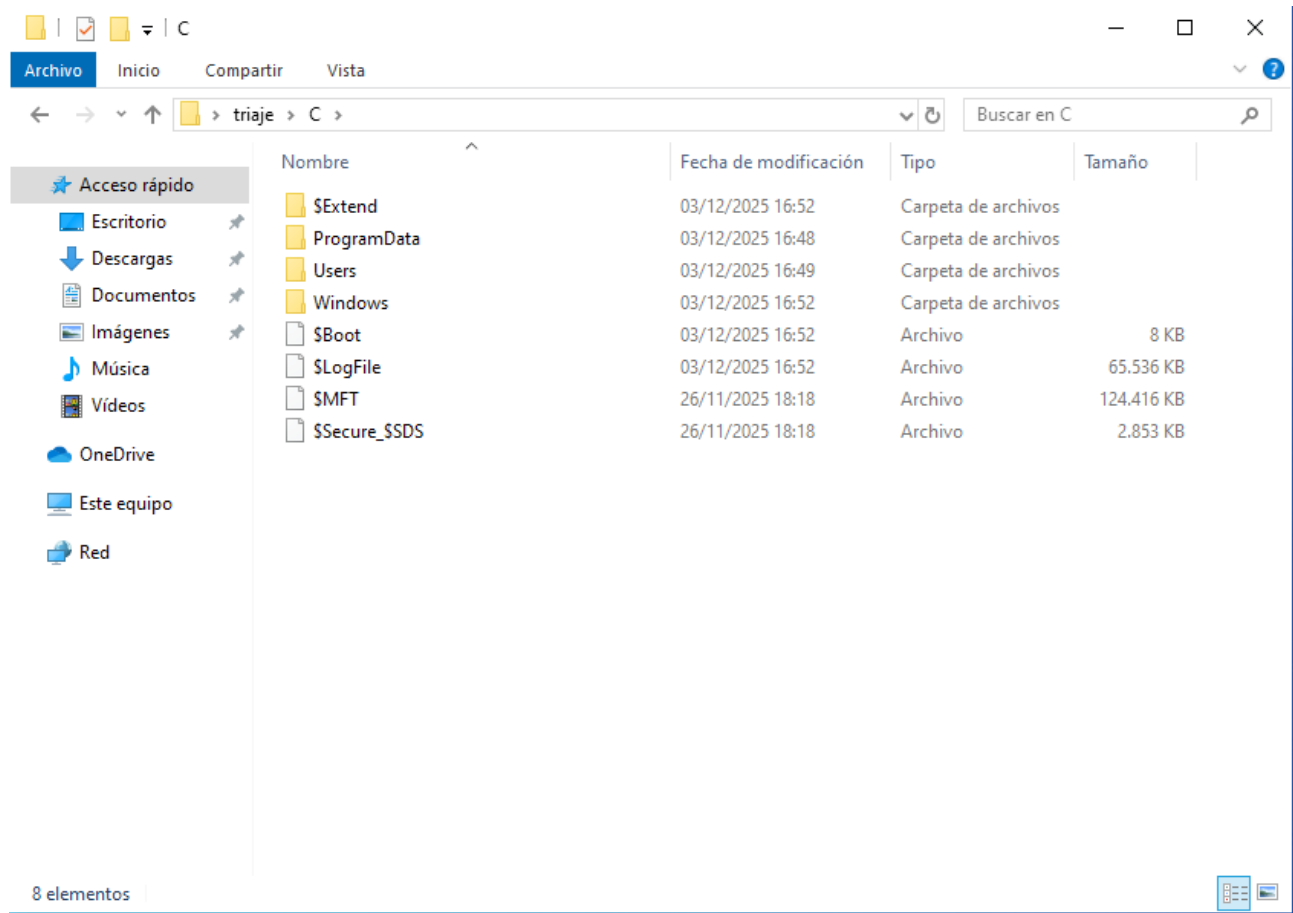
Copied 799 (Deduplicated: 207) out of 1,006 files in 453.4789 seconds. See C:\Users\Admin\Desktop\traje\2025-12-03T15_44
_32_5891800_CopyLog.csv for copy details

Total execution time: 453.6852 seconds

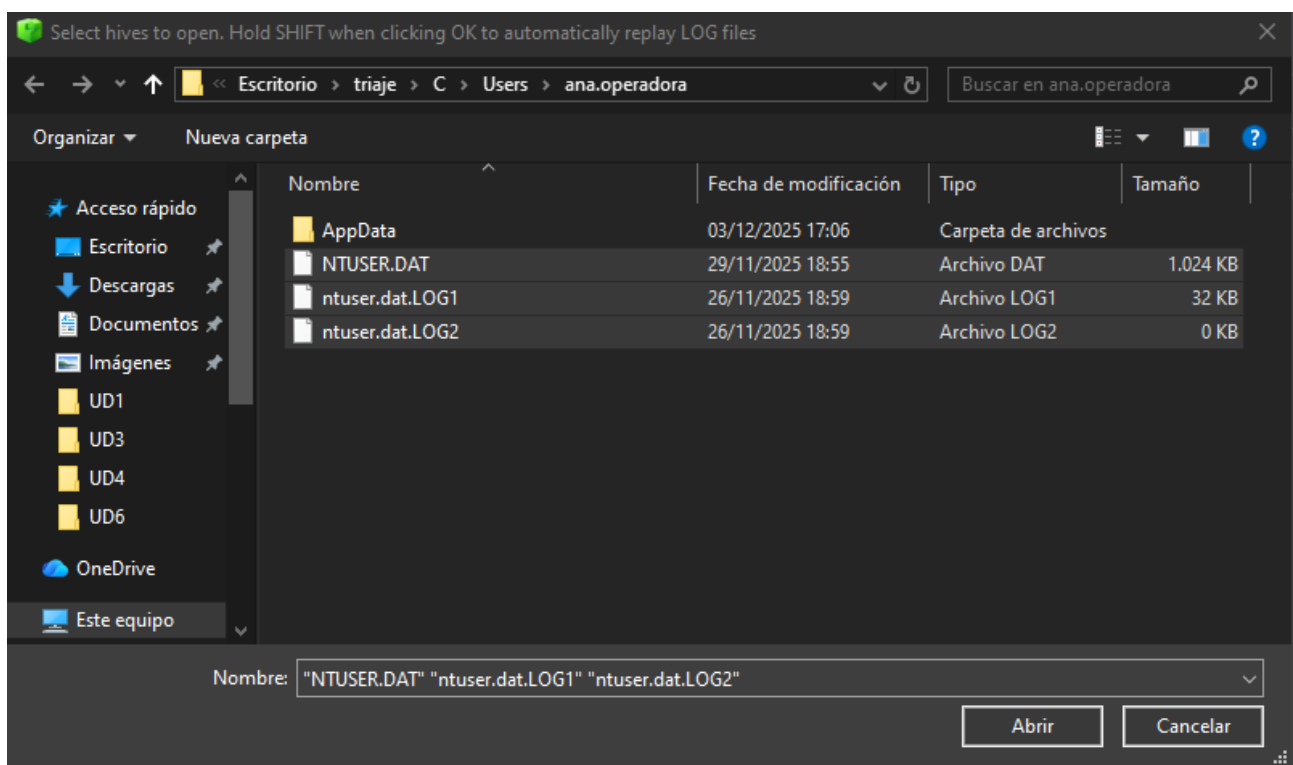
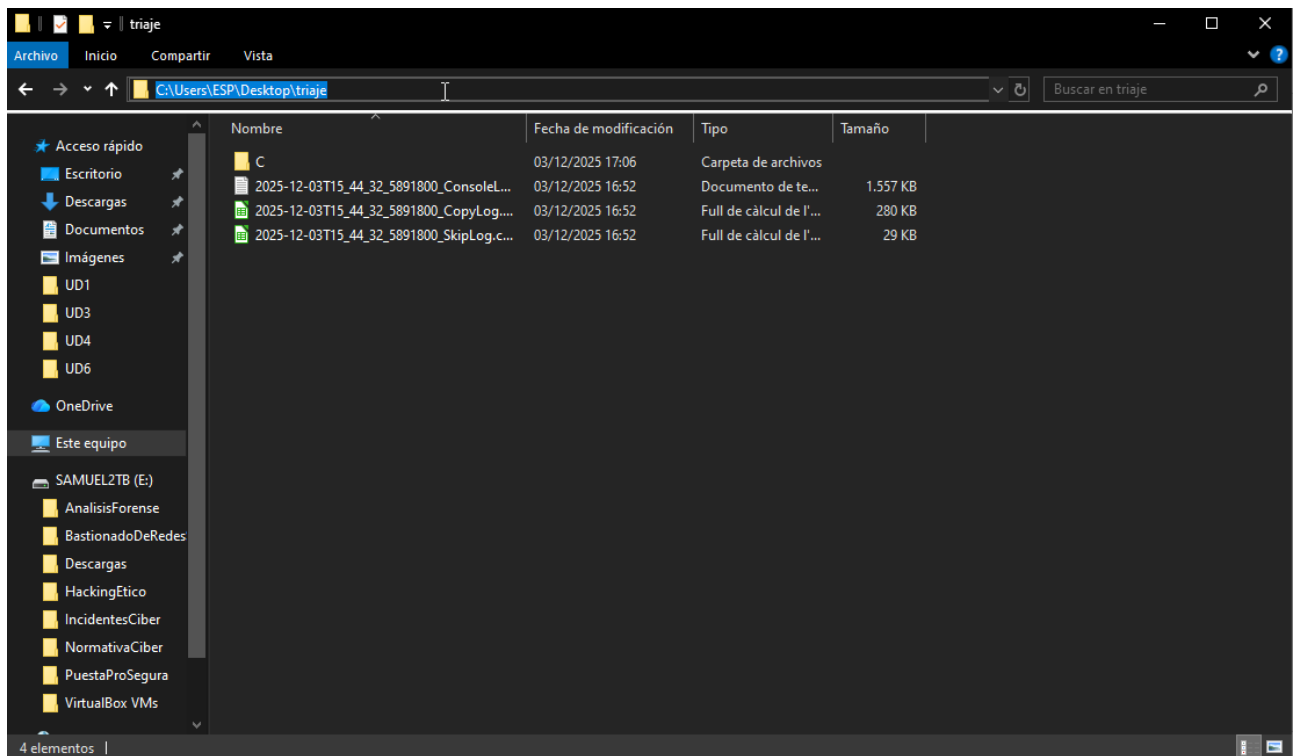
C:\Users\Admin\Desktop\kape>
```

Se ha escogido el triaje como método de adquisición ya que se disponía de poco tiempo para extraer los datos

2. Analisis del triaje



Una vez finalizado el triaje tenemos esta estructura de directorios, a continuación dejaríamos de usar la máquina virtual y y trabajaríamos en local



Analizaremos cada aplicación abierta por el usuario del sistema comenzando por ana.operadora, así para cada usuario

Registry Explorer v2.1.0

File Tools Options Bookmarks (30/0) View Help

Registry hives (1) Available bookmarks (30/0)

Enter text to search... Find

Key name # values # subkeys

Root

MountPoints2

Package Installation

RecentDocs

Ribbon

RunMRU

SearchPlatform

Shell Folders

Shutdown

StarPage

Streams

StuckRects3

Taskband

TypedPaths

User Shell Folders

UserAssist

VirtualDesktops

VisualEffects

Wallpapers

Ext

Feeds

FileAssociations

FileHistory

GameDVR

Group Policy

Holographic

ime

Values UserAssist

Drag a column header here to group by that column

Program Name	Run Counter	Focus Count	Focus Time	Last Executed
{System32}\notepad.exe	10	3	9 0d, 0h, 01m, 58s	2025-11-29 17:54:56
{Program Files x86}\FreeExcelViewer\freexcelviewer.exe	3	5	0d, 0h, 02m, 17s	2025-11-29 17:53:56
Microsoft.XboxGamingOverlay_8wekyb3d8bbwe!App	2	0	0d, 0h, 00m, 00s	2025-11-29 17:53:00
{User Pinned}\Taskbar\Microsoft Edge.lnk	1	0	0d, 0h, 00m, 00s	2025-11-28 07:48:51
MSEdge	1	1	0d, 0h, 01m, 06s	2025-11-28 07:48:51
C:\Users\Public\Desktop\Free Excel Viewer.lnk	1	0	0d, 0h, 00m, 00s	2025-11-28 07:29:27
C:\Users\ana.operatoradora\Desktop\FreeExcelViewer.exe	1	0	0d, 0h, 00m, 00s	2025-11-27 19:35:00
{User Pinned}\Taskbar\File Explorer.lnk	6	0	0d, 0h, 00m, 00s	2025-11-27 19:34:39
Microsoft.Windows.Explorer	6	5	0d, 0h, 01m, 58s	2025-11-27 19:34:39
{Programs}\System Tools\Command Prompt.lnk	1	0	0d, 0h, 00m, 00s	2025-11-27 19:32:48
{System32}\cmd.exe	1	3	0d, 0h, 01m, 09s	2025-11-27 19:32:48
{Programs}\Accessories\Notepad.lnk	6	0	0d, 0h, 00m, 00s	2025-11-26 17:57:32
{Common Programs}\Accessories\Paint.lnk	7	0	0d, 0h, 00m, 00s	2025-11-26 17:57:32
{Common Programs}\Accessories\Snipping Tool.lnk	9	0	0d, 0h, 00m, 00s	2025-11-26 17:57:32
{System32}\mspaint.exe	7	7	0d, 0h, 01m, 45s	2025-11-26 17:57:32
Microsoft.WindowsCalculator_8wekyb3d8bbwe!App	8	9	0d, 0h, 02m, 30s	2025-11-26 17:57:32

Total rows: 30

Export ?

Key: Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist Value: None Collapse all hives

ana.operatoradora

Registry Explorer v2.1.0

File Tools Options Bookmarks (29/0) View Help

Registry hives (1) Available bookmarks (29/0)

Enter text to search... Find

Key name # values # subkeys

Root

RecentDocs

Ribbon

RunMRU

SearchPlatform

Shell Folders

Shutdown

StarPage

StorageProvider

Streams

StuckRects3

Taskband

User Shell Folders

UserAssist

VirtualDesktops

VisualEffects

Wallpapers

PostBootReminders

Ext

Feeds

FileAssociations

FileHistory

GameDVR

Group Policy

Holographic

ime

ImmersiveShell

Values UserAssist

Drag a column header here to group by that column

Program Name	Run Counter	Focus Count	Focus Time	Last Executed
Microsoft.XboxGamingOverlay_8wekyb3d8bbwe!App	2	0	0d, 0h, 00m, 00s	2025-11-29 17:59:39
{System32}\notepad.exe	9	8	0d, 0h, 01m, 39s	2025-11-29 17:59:27
{Program Files x86}\FreeExcelViewer\freexcelviewer.exe	3	4	0d, 0h, 02m, 02s	2025-11-29 17:59:10
Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge	2	0	0d, 0h, 00m, 00s	2025-11-28 08:41:04
C:\Users\Public\Desktop\Free Excel Viewer.lnk	1	0	0d, 0h, 00m, 00s	2025-11-28 08:31:03
{User Pinned}\Taskbar\File Explorer.lnk	2	0	0d, 0h, 00m, 00s	2025-11-27 19:37:49
Microsoft.Windows.Explorer	2	1	0d, 0h, 00m, 08s	2025-11-27 19:37:49
{Programs}\System Tools\Command Prompt.lnk	1	0	0d, 0h, 00m, 00s	2025-11-27 19:37:05
{System32}\cmd.exe	1	1	0d, 0h, 00m, 43s	2025-11-27 19:37:05
{Programs}\Accessories\Notepad.lnk	6	0	0d, 0h, 00m, 00s	2025-11-26 19:13:14
{Common Programs}\Accessories\Paint.lnk	7	0	0d, 0h, 00m, 00s	2025-11-26 19:13:14
{Common Programs}\Accessories\Snipping Tool.lnk	9	0	0d, 0h, 00m, 00s	2025-11-26 19:13:14
{System32}\mspaint.exe	7	7	0d, 0h, 01m, 45s	2025-11-26 19:13:14
Microsoft.WindowsCalculator_8wekyb3d8bbwe!App	8	9	0d, 0h, 02m, 30s	2025-11-26 19:13:14
{System32}\SnippingTool.exe	9	11	0d, 0h, 03m, 15s	2025-11-26 19:13:14
Microsoft.MicrosoftStickyNotes_8wekyb3d8bbwe!App	10	13	0d, 0h, 04m, 00s	2025-11-26 19:13:14
Microsoft.People_8wekyb3d8bbwe!x4c	11	15	0d, 0h, 04m, 45s	2025-11-26 19:13:14

Total rows: 29

Export ?

Key: Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist Value: None Collapse all hives

Selected hive: NTUSER.DAT Last write: 2025-11-26 19:12:36 Key contains no values Load complete Hidden keys: 0 8

lucas.operatoradora

Registry Explorer v2.1.0

File Tools Options Bookmarks (29/0) View Help

Registry hives (1) Available bookmarks (29/0)

Enter text to search... Find

Key name

Values User Assist

Drag a column header here to group by that column

Program Name	Run Counter	Focus Count	Focus Time	Last Executed
(Program Files)FreeExcelViewer\FreeExcelViewer.exe		3	3 0d, 0h, 00m, 48s	2025-11-29 17:57:31
(User Pinned)TaskBar\File Explorer.lnk		8	0 0d, 0h, 00m, 00s	2025-11-29 17:57:14
Microsoft.Windows.Explorer		8	6 0d, 0h, 04m, 37s	2025-11-29 17:57:14
C:\Users\Public\Desktop\Microsoft Edge.lnk		1	0 0d, 0h, 00m, 00s	2025-11-29 17:55:50
MSEdge		3	4 0d, 0h, 05m, 01s	2025-11-29 17:55:50
(System32)\notepad.exe		10	10 0d, 0h, 03m, 43s	2025-11-28 09:05:26
(User Pinned)TaskBar\Microsoft Edge.lnk		2	0 0d, 0h, 00m, 00s	2025-11-28 09:02:24
Microsoft.XboxGamingOverlay_8wekyb3d8bbwe!App		1	0 0d, 0h, 00m, 00s	2025-11-28 08:44:17
(Programs)\System Tools\Command Prompt.lnk		3	0 0d, 0h, 00m, 00s	2025-11-27 19:42:08
(System32)\cmd.exe		3	3 0d, 0h, 01m, 00s	2025-11-27 19:42:08
(Program Files)\Windows NT\Accessories\notepad.exe		1	1 0d, 0h, 00m, 03s	2025-11-27 19:25:19
(Programs)\Accessories\Notepad.lnk		6	0 0d, 0h, 00m, 00s	2025-11-26 17:55:13
(Common Programs)\Accessories\Paint.lnk		7	0 0d, 0h, 00m, 00s	2025-11-26 17:55:13
(Common Programs)\Accessories\Snaping Tool.lnk		9	0 0d, 0h, 00m, 00s	2025-11-26 17:55:13
(System32)\mspaint.exe		7	7 0d, 0h, 01m, 45s	2025-11-26 17:55:13
Microsoft.WindowsCalculator_8wekyb3d8bbwe!App		8	9 0d, 0h, 02m, 30s	2025-11-26 17:55:13
(System32)\SnippingTool.exe		9	11 0d, 0h, 03m, 15s	2025-11-26 17:55:13

Total rows: 30 Export ?

Type viewer

Key: Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist Value: None Collapse all hives

Selected hive: NTUSER.DAT Last write: 2025-11-26 17:56:41 Key contains no values Load complete Hidden keys: 0 11

marta.operadora

Registry Explorer v2.1.0

File Tools Options Bookmarks (27/0) View Help

Registry hives (1) Available bookmarks (27/0)

Enter text to search... Find

Key name

Values User Assist

Drag a column header here to group by that column

Program Name	Run Counter	Focus Count	Focus Time	Last Executed
(Programs)\System Tools\Command Prompt.lnk		1	0 0d, 0h, 00m, 00s	2025-11-28 08:30:35
(System32)\cmd.exe		1	1 0d, 0h, 00m, 16s	2025-11-28 08:30:35
Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge		1	0 0d, 0h, 00m, 00s	2025-11-27 19:48:06
(Programs)\Accessories\Notepad.lnk		6	0 0d, 0h, 00m, 00s	2025-11-26 19:26:49
(Common Programs)\Accessories\Paint.lnk		7	0 0d, 0h, 00m, 00s	2025-11-26 19:26:49
(Common Programs)\Accessories\Snaping Tool.lnk		9	0 0d, 0h, 00m, 00s	2025-11-26 19:26:49
(System32)\notepad.exe		6	5 0d, 0h, 01m, 00s	2025-11-26 19:26:49
(System32)\mspaint.exe		7	7 0d, 0h, 01m, 45s	2025-11-26 19:26:49
Microsoft.WindowsCalculator_8wekyb3d8bbwe!App		8	9 0d, 0h, 02m, 30s	2025-11-26 19:26:49
(System32)\SnippingTool.exe		9	11 0d, 0h, 03m, 15s	2025-11-26 19:26:49
Microsoft.MicrosoftStickyNotes_8wekyb3d8bbwe!App		10	13 0d, 0h, 04m, 00s	2025-11-26 19:26:49
Microsoft.People_8wekyb3d8bbwe!x4c7a3b7d72188y46d4ya362y19ac5a5805e5x		11	15 0d, 0h, 04m, 45s	2025-11-26 19:26:49
Microsoft.WindowsMaps_8wekyb3d8bbwe!App		12	17 0d, 0h, 05m, 30s	2025-11-26 19:26:49
Microsoft.WindowsFeedbackHub_8wekyb3d8bbwe!App		13	19 0d, 0h, 06m, 15s	2025-11-26 19:26:49
Microsoft.Getstarted_8wekyb3d8bbwe!App		14	21 0d, 0h, 07m, 00s	2025-11-26 19:26:49
UEMEMCTLSESSION		23	0 0d, 0h, 00m, 00s	

Total rows: 23 Export ?

Type viewer

Key: Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist Value: None Collapse all hives

Selected hive: NTUSER.DAT Last write: 2025-11-26 19:23:14 Key contains no values Load complete Hidden keys: 0 14

soporte.it

Registry Explorer v2.1.0

File Tools Options Bookmarks (29/0) View Help

Registry hives (2) Available bookmarks (58/0)

Enter text to search... Find

Key name

Key: Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist

Value: None Collapse all hives

Selected hive: NTUSER.DAT_clean Last write: 2025-11-26 17:43:43 Key contains no values Load complete Hidden keys: 0 26

Program Name	Run Counter	Focus Count	Focus Time	Last Executed
(Programs)\System Tools\Command Prompt.Lnk	7	7	0 0d, 0h, 00m, 00s	2025-12-03 15:25:59
(System32)\cmd.exe	7	12	0d, 0h, 17m, 51s	2025-12-03 15:25:59
Y:\Vape.exe	1	0	0d, 0h, 00m, 00s	2025-12-03 15:25:48
Z:\MFTECmd\MFTECmd.exe	4	0	0d, 0h, 00m, 00s	2025-12-03 15:20:51
(User Pinned)\TaskBar\File Explorer.Lnk	6	0	0d, 0h, 00m, 00s	2025-12-03 15:14:36
Microsoft.Windows.Explorer	6	15	0d, 0h, 09m, 14s	2025-12-03 15:14:36
C:\Users\Public\Desktop\Free Excel Viewer.Lnk	1	0	0d, 0h, 00m, 00s	2025-12-03 15:11:47
(Program Files x86)\FreeExcelViewer\freexcelviewer.exe	1	2	0d, 0h, 00m, 39s	2025-12-03 15:11:47
Microsoft.XboxGamingOverlay_8wekyb3d8bbwe!App	1	0	0d, 0h, 00m, 00s	2025-12-03 15:05:02
D:\XboxWindowsAdditions-amd64.exe	1	4	0d, 0h, 00m, 36s	2025-11-26 17:48:05
(Programs)\Accessories\Notepad.Lnk	6	0	0d, 0h, 00m, 00s	2025-11-26 17:42:03
(Common Programs)\Accessories\Paint.Lnk	7	0	0d, 0h, 00m, 00s	2025-11-26 17:42:03
(Common Programs)\Accessories\Snipping Tool.Lnk	9	0	0d, 0h, 00m, 00s	2025-11-26 17:42:03
(System32)\notepad.exe	6	5	0d, 0h, 01m, 00s	2025-11-26 17:42:03
(System32)\mspaint.exe	7	7	0d, 0h, 01m, 45s	2025-11-26 17:42:03
Microsoft.WindowsCalculator_8wekyb3d8bbwe!App	8	9	0d, 0h, 02m, 30s	2025-11-26 17:42:03
(System32)\SnippingTool.exe	9	11	0d, 0h, 03m, 15s	2025-11-26 17:42:03
Microsoft.MicrosoftStickyNotes_8wekyb3d8bbwe!App	10	13	0d, 0h, 04m, 00s	2025-11-26 17:42:03

Total rows: 38 Export ?

admin

3. Extracción de la MFT

```

C:\Windows\system32>cd C:\Users\ESP\Desktop\MFTECmd

C:\Users\ESP\Desktop\MFTECmd>MFTECmd.exe -f "C:\Users\ESP\Desktop\triaje\C\MFT" --body . --bodyf examen.body --bd1 C
MFTECmd version 1.3.0.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/MFTECmd

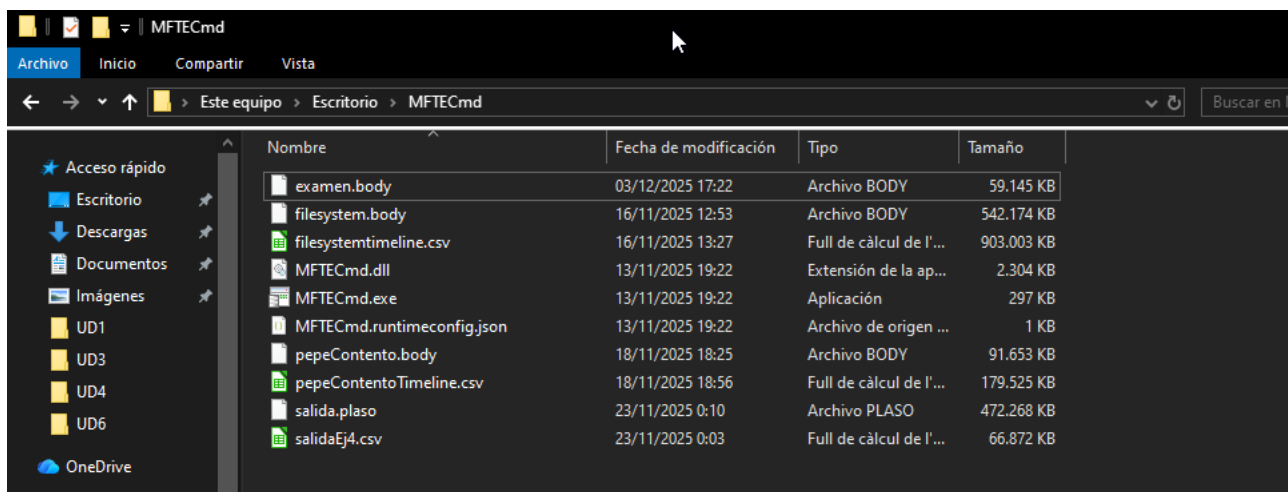
Command line: -f C:\Users\ESP\Desktop\triaje\C\MFT --body . --bodyf examen.body --bd1 C
File type: Mft

Processed C:\Users\ESP\Desktop\triaje\C\MFT in 3,5022 seconds

C:\Users\ESP\Desktop\triaje\C\MFT: FILE records found: 124,408 (Free records: 0) File size: 121,5MB
Bodyfile output will be saved to .\examen.body

C:\Users\ESP\Desktop\MFTECmd>

```



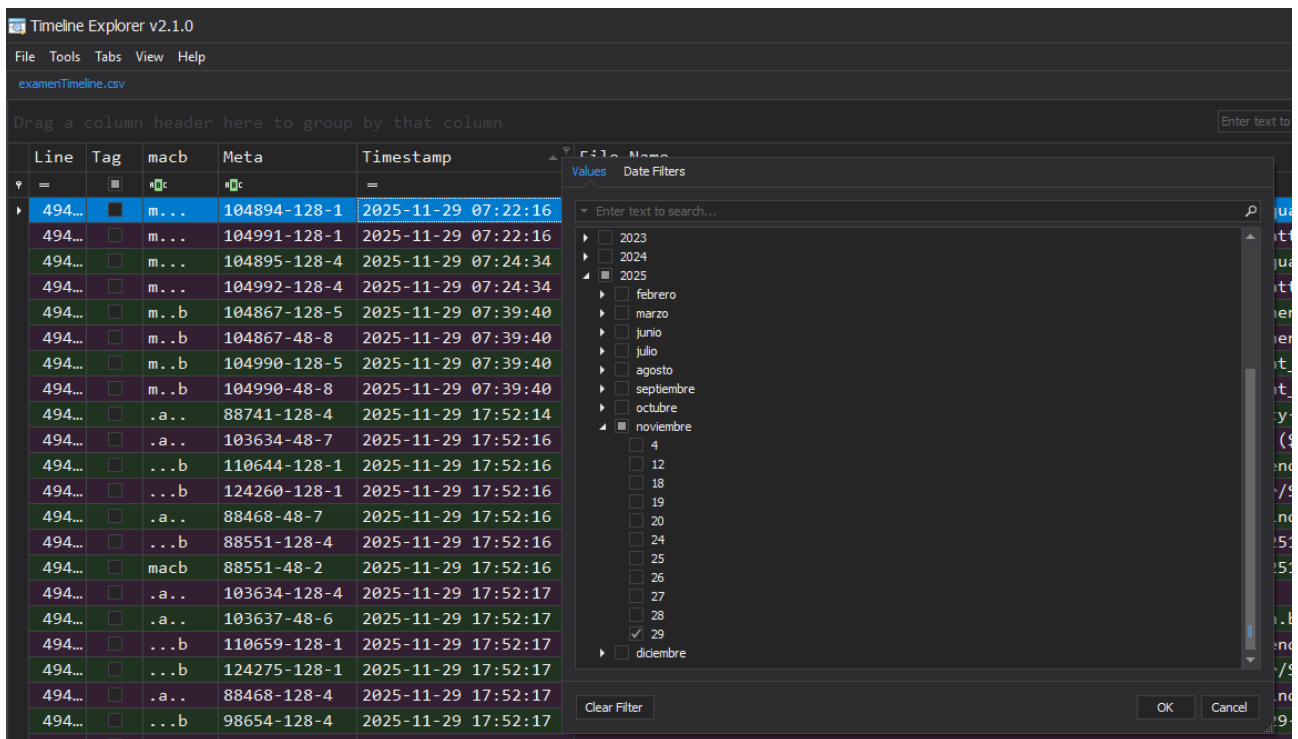
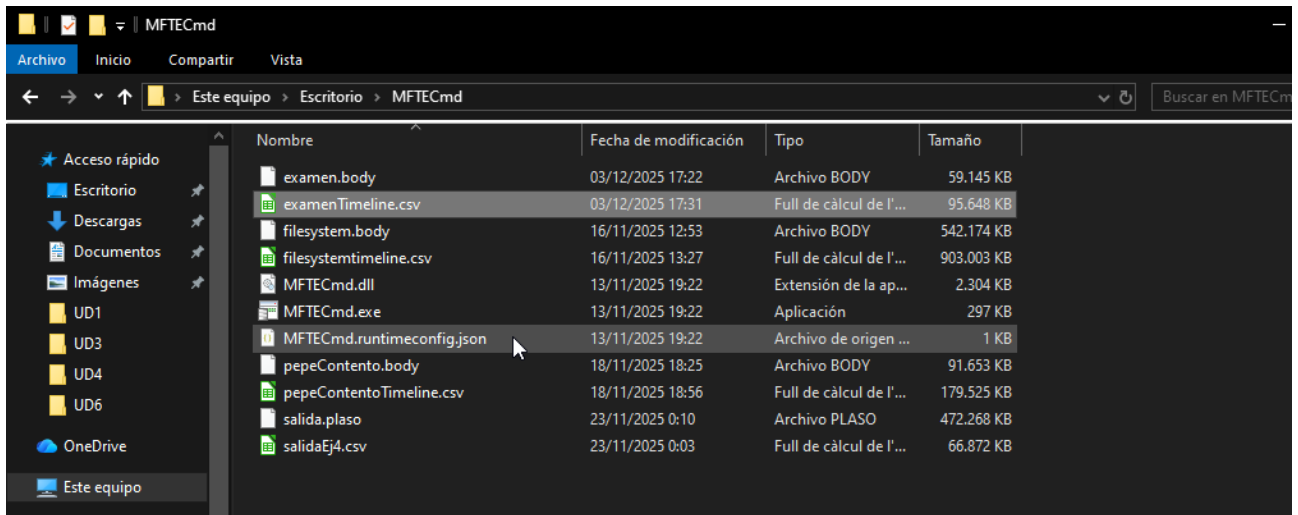
```
(samuel@kali)-[~]
$ cd /media/sf_MFTECmd

(samuel@kali)-[/media/sf_MFTECmd]
$ ls
examen.body      filesystemtimeline.csv
filesystem.body  MFTECmd.dll

(samuel@kali)-[/media/sf_MFTECmd]
$
```

```
(samuel@kali)-[/media/sf_MFTECmd]
$ mactime -z UTC -y -d -b examen.body > ./examenTimeline.csv
Old package separator "'" deprecated at /usr/bin/mactime line 154.
Old package separator "'" deprecated at /usr/bin/mactime line 167.

(samuel@kali)-[/media/sf_MFTECmd]
$
```



Timeline Explorer v2.1.0

File Tools Tabs View Help

examenTimeline.csv

Drag a column header here to group by that column

clientes

Find

Line	Tag	macb	Meta	Timestamp	File Name
4996...	m.c.	67807-128-1	2025-11-29 17:53:56	c:/Users/ana.operadora/AppData/Roaming/Microsoft/Windows/Recent/clientes.lnk	
4996...	mac.	67807-48-2	2025-11-29 17:53:56	c:/Users/ana.operadora/AppData/Roaming/Microsoft/Windows/Recent/clientes.lnk (\$F	
4996...	ma..	110946-128-3	2025-11-29 17:53:56	c:/Users/ESP/Desktop/triajeExamen/C/Users/ana.operadora/AppData/Roaming/Microsof	
5018...	ma..	110962-128-3	2025-11-29 17:59:10	c:/Users/ESP/Desktop/triajeExamen/C/Users/lucas.operadora/AppData/Roaming/Micros	
5012...	ma..	110978-128-1	2025-11-29 17:57:31	c:/Users/ESP/Desktop/triajeExamen/C/Users/marta.operadora/AppData/Roaming/Micros	
5018...	m.c.	1908-128-1	2025-11-29 17:59:10	c:/Users/Lucas.operadora/AppData/Roaming/Microsoft/Windows/Recent/clientes.lnk	
5018...	mac.	1908-48-2	2025-11-29 17:59:10	c:/Users/lucas.operadora/AppData/Roaming/Microsoft/Windows/Recent/clientes.lnk (	
5012...	m.c.	103644-128-1	2025-11-29 17:57:31	c:/Users/marta.operadora/AppData/Roaming/Microsoft/Windows/Recent/clientes.lnk	
5012...	mac.	103644-48-2	2025-11-29 17:57:31	c:/Users/marta.operadora/AppData/Roaming/Microsoft/Windows/Recent/clientes.lnk (	

Como conclusion marta.operadora ha sido la culpable de extraviar el fichero pues es la ultima que lo tocó según el timestamp